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Application of Schwarz–Christoffel mapping to the analysis of conduction through a slot

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We consider the generic problem of steady conduction through a slot traversing a non-conducting plate that separates two semi-infinite conducting regions. The current-density field is conservative; the dimensionless problem governing its potential depends upon a single geometric parameter, h, the ratio of the slot length (i.e. the plate thickness) to its width. We construct a Schwarz–Christoffel transformation to handle this two-dimensional transport problem. The transformation is expressed in terms of two parameters which are related to h through two implicit equations; in the limit $h \to 0$, it becomes explicit. Because of the slow decay of the current density at large distances from the slot, the integral representing the slot resistance diverges. The excess resistance of a finite-length slot relative to that of a zero-length slot is, however, finite. This excess resistance depends only upon the asymptotic behaviour of the potential far from the slot; it may therefore be directly obtained as a function of the two transformation parameters. Asymptotic approximations are found for the excess resistance at small and large h, respectively, scaling as $h \ln h$ and h. The single-slot solution is used to analyse conduction through a periodic array of widely spaced slots.

1. Introduction

The class of conduction-like problems, in which a field satisfies the Laplace equation with homogeneous Neumann boundary conditions, includes a wide range of

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physical phenomena such as Ohmic conduction of charge [1], inviscid irrotational flows [2], Hele–Shaw flows [3], heat and mass transport at zero Péclet numbers [4] and frictional slip of longitudinal shear flows at superhydrophobic surfaces [5].

We investigate here the classical problem of the resistance to conduction through a pore traversing an infinite insulating plate that separates two semi-infinite conducting regions. For convenience, we present our analysis using the terminology of Ohmic conduction, using σ to denote the electrical conductivity of the conducting medium.

The axisymmetric problem of conduction through a circular hole of radius a in a zero-thickness insulating plate may be solved analytically by means of separation of variables in oblate spheroidal coordinates [6]. While this method cannot be used to analyse conduction through a pore of non-zero length, physical intuition suggests that the resistance of a pore of uniform cross-section traversing an insulating plate of thickness $2ha > 0$ can be approximated by the sum of exit and entrance resistances (i.e. the resistance of a hole in a plate of zero thickness), together with the intrinsic resistance of the pore itself, based upon its length and the resistance per unit length of an infinite cylinder of the same radius. Indeed, Sherwood *et al.* [7] showed via numerical computation that the resistance of the system can be written as

$$\frac{1}{a\sigma}\left(\frac{1}{2}+\frac{2h}{\pi}+\theta\right), \tag{1.1}$$

where the first term represents the combined entrance and exit resistance, the second represents the intrinsic resistance of a uniform circular cylinder, and the third term is a correction, with $\theta = 0$ when $h = 0$ and $\theta \to 0.229$ in the limit $h \to \infty$.[1]

Here, we consider the comparable two-dimensional problem of planar conduction through an infinite slot traversing an insulating plate. This two-dimensional problem is motivated by the flow of both liquids and compressible gases through shallow micro-channels [9], which are commonly employed in microfluidic devices [10]. Not only are these two-dimensional problems of interest in their own right, but they are also of interest for comparison with similar problems in three dimensions for which no exact results are available.

This problem has been previously solved, using conformal-mapping techniques, in various physical contexts [11–13]. The results, however, typically appear in a form that is difficult to understand (e.g. [12, eqns (2.16)–(2.17) and fig. 5]). Here, we present (in §3) a simplified derivation of a conformal-mapping solution, which we then use (in §4) to calculate the slot resistance. In particular, we derive useful closed-form approximations for plates of either small or large thickness. Our single-slot solution is then used (in §5) to determine the effective resistance of an array of slots, thereby allowing comparison with previously published solutions of the classical array problem.

2. Problem formulation

We consider a two-dimensional conduction problem, investigating the electrical resistance of a slot of width $2a$ in an insulating plate of thickness $2ha$. We adopt Cartesian coordinates (ax, ay) with the x-axis along the centre-line of the plate thickness and the y-axis perpendicular to the plate. The geometry and coordinates are shown in figure 1.

A current of magnitude I (per unit length in the direction normal to the xy-plane) flows through the slot from the upper region $y > 0$ to the lower region $y < 0$. We non-dimensionalize all length variables by a and the current density $\mathbf{i}$ by I/a. In steady state, charge conservation requires that

$$\nabla \cdot \mathbf{i} = 0. \tag{2.1}$$

[1]Similarly, the hydraulic resistance for Stokes flow through a cylindrical hole in a plate was studied by Dagan *et al.* [8] and could be expressed as the sum of an entrance and exit resistance, the hydraulic resistance of a uniform cylindrical pipe, and a small correction that was found by numerical computation to tend to a constant in the thick-plate limit.

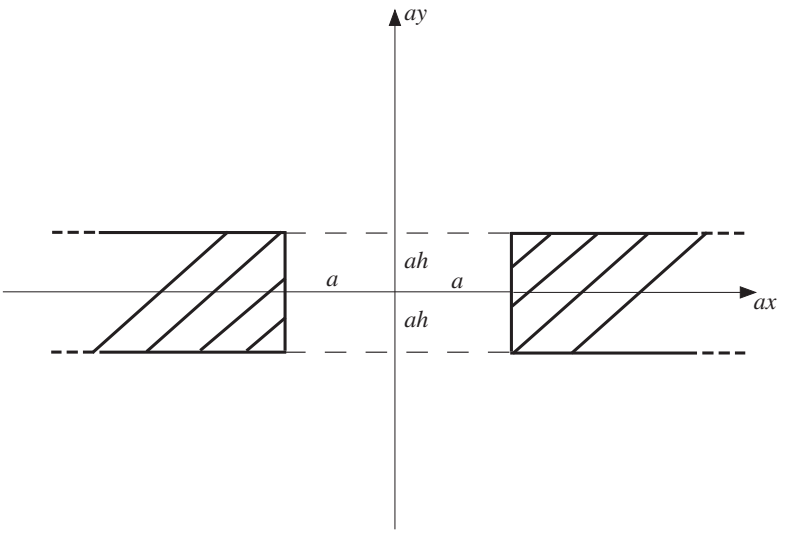


Figure 1. An insulating plate of thickness $2ha$ perforated by a slot of width $2a$. The plate is surrounded by a medium with electrical conductivity σ, and a current I flows through the slot from infinity in the upper half-plane to infinity in the lower half-plane.

As the solid walls of the plate are non-conducting, the current density satisfies the boundary condition

$$\hat{\mathbf{n}} \cdot \mathbf{i} = 0, \tag{2.2}$$

where $\hat{\mathbf{n}}$ is the unit vector normal to the wall. Far from the slot, where $r = \sqrt{x^2 + y^2} \gg 1$,

$$\mathbf{i} \sim \mp \frac{\hat{\mathbf{e}}_\mathrm{r}}{\pi r} \quad \text{as } r \to \infty, \tag{2.3}$$

where $\hat{\mathbf{e}}_\mathrm{r}$ is a unit vector in the radial direction and the sign on the right-hand side is set according as $y \gtrless 0$.

We assume Ohmic conduction with uniform conductivity σ, so that

$$\mathbf{i} = -\boldsymbol{\nabla}\varphi, \tag{2.4}$$

where the potential φ has been normalized by I/σ. Equation (2.1) for charge conservation then implies that the electric potential φ is harmonic, while (2.2) becomes the homogeneous Neumann condition

$$\frac{\partial \varphi}{\partial n} = 0, \tag{2.5}$$

where $\partial/\partial n = \hat{\mathbf{n}} \cdot \boldsymbol{\nabla}$ is the derivative normal to the plate boundary. In the far field, (2.3)–(2.4) yield

$$\varphi \sim \pm \frac{1}{\pi} \ln r \quad \text{as } r \to \infty. \tag{2.6}$$

Several symmetry properties are readily deduced. Antisymmetry in the y-direction implies that

$$\varphi = 0 \quad \text{at } y = 0, \tag{2.7}$$

while symmetry about $x = 0$ implies that

$$\frac{\partial \varphi}{\partial x} = 0 \quad \text{at } x = 0. \tag{2.8}$$

In what follows, we exploit the latter symmetry and consider only the transport in the domain $\mathcal{D}$ which consists of the conducting region in the half-plane $x > 0$. Its boundary $\partial\mathcal{D}$ thus consists of the genuine boundary of the right-hand half-plate together with the y-axis. Condition (2.8) then implies that (2.5) applies on the whole $\partial\mathcal{D}$.

3. Conformal-mapping solution

To calculate the harmonic function $\varphi(x, y)$, we regard it as the real part of a complex function $F(z)$ of the complex variable $z = x + \mathrm{i}y$:

$$\varphi(x, y) = \Re\{F(z)\}. \tag{3.1}$$

The far-field condition (2.6) is then transformed to

$$F(z) \sim \pm\frac{1}{\pi}\ln z \quad \text{as } r \to \infty, \tag{3.2}$$

with the sign set according as $y \gtrless 0$.

We now consider a conformal mapping $\zeta = g(z)$ that takes the domain $\mathcal{D}$ in the z-plane onto the upper half-space $\eta > 0$ of the complex $\zeta = \xi + \mathrm{i}\eta$ plane. It is well known from the theory of conformal mappings [14] that if $f(\zeta)$ is analytic in the half-plane, with the real part satisfying a homogeneous Neumann boundary condition on the image of $\partial\mathcal{D}$, then the real part of $f[g(z)]$ is harmonic in $\mathcal{D}$ and satisfies a homogeneous Neumann condition on $\partial\mathcal{D}$. Moreover, if $f[g(z)]$ also satisfies (3.2), then it follows that the requisite complex potential is

$$F(z) = f[g(z)]. \tag{3.3}$$

By the symmetry of the problem, we can impose that the upper (EE') and lower (AA') infinities in the z-plane (figure 2) are, respectively, mapped to $\zeta = \pm 1$. The calculation of $f(\zeta)$ is then straightforward, as the far-field condition (3.2) represents a net current of magnitude $\frac{1}{2}$ arriving from EE' and leaving through AA': we therefore need to specify a source of unit strength[2] at $\zeta = 1$ and a sink of unit strength at $\zeta = -1$:

$$f(\zeta) \sim \pm\frac{1}{2\pi}\ln(\zeta \pm 1) \quad \text{as } \zeta \to \pm 1. \tag{3.4}$$

It is readily verified that the function that satisfies (3.4) with real part satisfying a homogeneous Neumann boundary condition on the ξ-axis (the image of $\partial\mathcal{D}$) is simply

$$f(\zeta) = \frac{1}{2\pi}\ln\frac{\zeta + 1}{\zeta - 1}. \tag{3.5}$$

(a) Schwarz–Christoffel mapping

It remains for us to find the mapping $\zeta = g(z)$. Since $\mathcal{D}$ is a degenerate polygon whose boundary $\partial\mathcal{D}$ is $ABCDEE'OA'$ (figure 2), g may be constructed as a Schwarz–Christoffel transformation. Using the problem symmetry, we choose a transformation from z to ζ that maps AA' to -1, B to $-k$, D to k and EE' to 1 (figure 2). The transformation thus satisfies [14]

$$\frac{\mathrm{d}z}{\mathrm{d}\zeta} = -\mathrm{i}M\frac{(\zeta^2 - k^2)^{1/2}}{(\zeta^2 - 1)^{3/2}}. \tag{3.6}$$

By symmetry, C is mapped to 0, so integration yields

$$z = 1 - \mathrm{i}M\int_0^\zeta \frac{(t^2 - k^2)^{1/2}}{(t^2 - 1)^{3/2}}\,\mathrm{d}t, \tag{3.7}$$

[2]Note that only half of the current emanating from that source goes into the upper half of the ζ-plane.

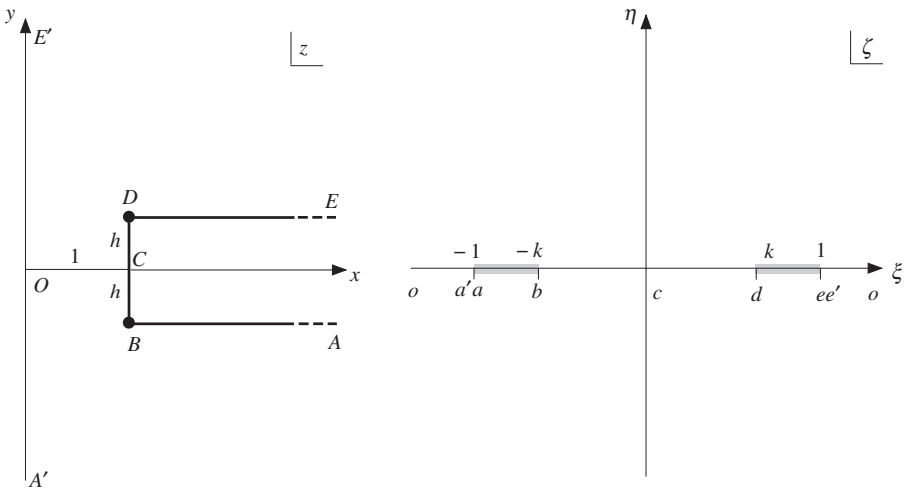


Figure 2. The physical (z) plane and the transformed (ζ) plane.

which provides g implicitly. The branch of $(\zeta^2-k^2)^{1/2}/(\zeta^2-1)^{3/2}$ in (3.6) and (3.7) is selected as that defined by the branch cuts in figure 2, together with the requirement of being real and negative on the upper η-axis.

The transformation parameters k and M are set by the requirement that $\mathcal{D}$ is indeed mapped to the upper half of the ζ-plane. The correct angles at the vertices of $\partial\mathcal{D}$ are already ensured by the transformation (3.7); it therefore remains to fix two lengths, namely the non-dimensional channel half-width (unity) and the channel half-length h. The first condition, that the slot has unit half-width, is satisfied by requiring that the origin O of the z-plane be mapped to o, the point at infinity in the ζ-plane. It is convenient to impose this condition on the point $\zeta=\mathrm{i}\infty$, so that in (3.7) $t=\mathrm{i}\eta$ with $0<\eta<\infty$. With the above-mentioned choice of branch in the integrand of (3.7), this condition leads to

$$M\int_0^\infty \frac{(\eta^2+k^2)^{1/2}}{(\eta^2+1)^{3/2}}\,\mathrm{d}\eta=1, \tag{3.8}$$

where the integral is real valued. This equation may be written as

$$M\mathrm{E}(\sqrt{1-k^2})=1, \tag{3.9}$$

where

$$\mathrm{E}(m)=\int_0^1\sqrt{\frac{1-m^2t^2}{1-t^2}}\,\mathrm{d}t \tag{3.10}$$

is the complete elliptic integral of the second kind [15].

The requirement that the channel has half-length h implies that point D is mapped to k. Substituting into (3.7), where now $t=\xi$ with $\xi>0$, thus yields

$$M\int_0^k \frac{(k^2-\xi^2)^{1/2}}{(1-\xi^2)^{3/2}}\,\mathrm{d}\xi=h. \tag{3.11}$$

Since the integrals appearing in both (3.8) and (3.11) are real-valued and positive, these conditions determine M and k as positive real numbers. That M is real and positive was to be expected: with our choice of branch in (3.6), $(\zeta^2-k^2)^{1/2}/(\zeta^2-1)^{3/2}$ is real and negative on bcd; this segment, on which $\mathrm{d}\zeta=\mathrm{d}\xi$, is transformed by (3.7) onto the segment BCD, on which $\mathrm{d}\zeta=\mathrm{i}\,\mathrm{d}y$. (This was the reason for writing the scaling factor in (3.6) as $-\mathrm{i}M$.)

It is worth noting that transformation (3.7) may be expressed in terms of the Jacobi elliptic functions (cf. [16]). Our explicit approach, wherein the branch cuts are clearly identified, is more convenient for the present problem, as it allows us to obtain the parameters M and k in terms of the real-valued integrals in (3.8) and (3.11).

(b) Limiting cases

Further analytic progress can be made in two cases: (i) the case $h \ll 1$, when the plate is thin compared with the width of the slot, and (ii) the limit $h \gg 1$, when the plate is thick.

(i) A thin plate, $h \ll 1$

Consider first the 'slit' limit of zero thickness. In this limit, D approaches C in the z-plane, implying that $k \to 0^+$ (so that d approaches c in the ζ-plane). Since $\mathrm{E}(1) = 1$, we see from (3.9) that $M = 1$. The transformation (3.7) can thus be written as

$$z = \frac{\mathrm{i}}{(\zeta^2 - 1)^{1/2}}, \tag{3.12}$$

which may be readily inverted to give $\zeta = g(z)$ explicitly (cf. [17]):

$$g(z) = \frac{(z^2 - 1)^{1/2}}{z}. \tag{3.13}$$

The branch in (3.13) is chosen so that ζ is real and positive when z is on the upper y-axis. This maps the lower half-plane infinity (AA') to -1 and the upper half-plane infinity (CC') to 1.

Consider now the more delicate limit of small but finite h, $0 < h \ll 1$. In this limit, we need to know how k and M deviate from zero and unity, respectively. We set $\xi = k\tau$ ($k \ll 1$) in (3.11) so that at leading order

$$k^2 \int_0^1 (1 - \tau^2)^{1/2} \, \mathrm{d}\tau \sim h. \tag{3.14}$$

The integral in (3.14) is $\pi/4$, hence

$$k \sim \left(\frac{4h}{\pi}\right)^{1/2} + \cdots. \tag{3.15}$$

To determine M, we employ (3.9), together with the asymptotic expansion [15]

$$\mathrm{E}(t) = 1 + (1 - t)[\tfrac{3}{2} \ln 2 - \tfrac{1}{2} \ln(1 - t) - \tfrac{1}{2}] + o(1 - t) \quad \text{for } 1 - t \ll 1, \tag{3.16}$$

to obtain

$$M = 1 - \frac{2h}{\pi}\left(\ln 4 - \frac{1}{2} \ln \frac{4h}{\pi} - \frac{1}{2}\right) + o(h). \tag{3.17}$$

(ii) A thick plate, $h \gg 1$

When the slot half-length h is large, we expect that $k \to 1^-$. Since $\mathrm{E}(t) = \pi/2 + O(t^2)$ for small t [15], we find from (3.9)

$$M = \frac{2}{\pi} + O(1 - k). \tag{3.18}$$

To obtain a leading-order approximation to $1 - k$, we evaluate the integral in (3.11) by splitting the integration domain [18] to obtain

$$\tfrac{3}{2} \ln 2 - 1 - \tfrac{1}{2} \ln(1 - k) + o(1). \tag{3.19}$$

Equation (3.11) becomes

$$\left[\frac{2}{\pi}+O(1-k)\right]\left[\frac{3}{2}\ln 2-1-\frac{1}{2}\ln(1-k)+o(1)\right]=h. \tag{3.20}$$

It follows that

$$1-k\sim\frac{8}{\mathrm{e}^2}\,\mathrm{e}^{-\pi h}. \tag{3.21}$$

4. The slot resistance

In principle, we can now use (3.1) and (3.3) to calculate the potential distribution for any given value of h. The complex potential $f(\zeta)$ is given by (3.5) and the mapping $g(z)$ is provided by (3.7), with k and M determined by the simultaneous solution of (3.9) and (3.11). Since $g(z)$ is known only implicitly, the potential would be obtained as a collection of numerical values, rather than in closed form. In many cases, however, our interest lies not in the detailed potential distribution but rather in integral properties, whose dependence on h may be sought analytically. In the present problem, the key quantity of interest is the slot resistance.

Consider an arc of radius $r\gg 1$, in the upper-half z-plane, which is centred at the origin. We anticipate that, as r becomes large and $o(1)$ terms are neglected, the potential along that arc becomes uniform, i.e. independent of the argument θ of z (see (4.5)). The potential at the mid-plane $y=0$ across the centre of the slot is also uniform (see (2.7)). We can therefore define the Ohmic resistance (for a unit length in the direction perpendicular to the xy-plane) of the region bounded between the arc and mid-plane: normalized by $1/\sigma$, it is simply given by the value of φ on the arc. (Recall the φ has been normalized by I/σ, where I is the current.)

Because of the slow divergence of φ with r (2.6), we cannot define the resistance of the entire system.[3] Noting, however, that this divergence is independent of h, we can define the *excess* resistance of a finite-length slot relative to that of a slit ($h=0$). Denoting the electric potential in the slit limit by φ_0, this excess resistance is given by

$$R=\lim_{r\to\infty}(\varphi-\varphi_0). \tag{4.1}$$

To obtain R, we start with the calculation of φ_0. In the slit limit $h\to 0$, the transformation $\zeta=g(z)$ is provided explicitly by (3.13). The associated potential is readily obtained from (3.1), (3.3) and (3.5):

$$\varphi_0=\frac{1}{2\pi}\Re\{\ln[(1-z^{-2})^{1/2}+1]-\ln[(1-z^{-2})^{1/2}-1]\}. \tag{4.2}$$

Noting the branch for which (3.13) is defined, this potential has the far-field form[4]

$$\varphi_0=\frac{1}{\pi}[\ln r+\ln 2+O(r^{-2})]. \tag{4.3}$$

For non-zero thickness, $h>0$, the mapping $g(z)$ between ζ and z is provided implicitly by (3.7), with M and k given by the solution of (3.9) and (3.11). Accordingly, we cannot obtain an analytic expression for φ. Fortunately, the determination of the excess resistance R using (4.1) requires only

[3]Unlike the equivalent axisymmetric problem of a circular hole in a plate, for which the current far from the hole decays as r^{-2} and the potential varies as r^{-1}.

[4]If a circular perfect conductor of unit radius were to be placed at the origin, the potential in the remainder of the half-plane (namely $r>1$) would be a function of r alone, namely $\pi^{-1}\ln r$; the resistance between $r=1$ and the arc at $r>1$ would then be given by the first term of (4.3). The term $\ln 2$ in (4.3) thus represents an additional resistance of the entry region into the slit.

knowledge of φ at large $|z|$ in the upper half-plane, corresponding to $1-\zeta \ll 1$. The dominant contribution to (3.7) in this limit is a local one [18], concentrated about $t=\zeta$, and we readily find

$$z = \frac{M(1-k^2)^{1/2}}{2^{1/2}(1-\zeta)^{1/2}} + O(1). \tag{4.4}$$

Inverting (4.4) thus provides an explicit approximation for $g(z)$ that may be inserted into (3.5). Equations (3.3) and (3.1) then yield

$$\varphi = \frac{1}{\pi}\left[\ln r + \ln 2 - \ln M - \frac{1}{2}\ln(1-k^2) + O(r^{-1})\right]. \tag{4.5}$$

Substitution of the asymptotic approximations (4.3) and (4.5) into (4.1) gives

$$R = -\frac{1}{2\pi}[2\ln M + \ln(1-k^2)]. \tag{4.6}$$

Since M and k are provided by (3.9) and (3.11), we have obtained R as an implicit function of h.

We now consider the two asymptotic limits discussed in §3b. For $h \ll 1$, substitution of (3.15) and (3.17) into (4.6) yields

$$R = \frac{h}{\pi^2}\left(\ln\frac{4\pi}{h} + 1\right) + o(h). \tag{4.7}$$

For $h \gg 1$ (thick plate), $1-k$ is given by (3.21), with $1+k \sim 2$ and $M \sim 2/\pi$ to within an exponentially small error (3.18). The excess resistance is therefore

$$R = \frac{h}{2} + \frac{1}{\pi}\left(1 - \ln\frac{8}{\pi}\right) + o(1) = \frac{h}{2} + 0.02078 + o(1). \tag{4.8}$$

Note that the first term on the right-hand side of (4.8) represents the intrinsic resistance of the upper half of the slot (namely a rectangle of length h and unit half-width): as h becomes large, this term dominates the excess resistance. Physically, an excess resistance of exactly $R = h/2$ could be realized by introducing perfectly conducting planes at $y = \pm h$, across the pore entrance and exit.[5] Removing these conductors can only increase the system resistance, hence the simple estimate $R = h/2$ must be a lower bound to the true resistance, as is seen in (4.8). The asymptote (4.8) for $h \gg 1$ has the same form as the numerically derived approximation (1.1) that describes the resistance of an axisymmetric cylindrical pore, for which no analytic result is available.

More generally, we can determine k and M for arbitrary values of h by the simultaneous solution of (3.9) and (3.11). The variation of R with h, as given by (4.6), is displayed in figure 3. Also shown are the small-h and large-h asymptotes (4.7) and (4.8). The asymptote (4.8) is in error by less than 1% for $h > 0.31$. The differences between the results for R computed from (4.6) and the asymptotes (4.7) and (4.8) are more easily seen in figure 4, which shows $R - h/2$ as a function of h.

5. An array of slots

We now address the relation of our results to the problem of conduction through a periodically perforated plate. The dimensional geometry, described in figure 5, now consists of an infinite array of slots (each of width $2a$ and length $2ah$) spaced regularly, with the nth slot ($-\infty < n < \infty$) centred about the point $(2naL, 0)$. Rather than specifying a net current (per unit length in the direction normal to the xy-plane), we now impose an average current density of magnitude $(2aL)^{-1}I$ in the negative y-direction, so that the net current (per unit length in the direction normal to the xy-plane) passing through each slot is I, as in the single-slot problem. We employ the same

[5]When such planes are inserted, transport outside the slot is identical to that outside a slit and is independent of h, whereas that within the slot is one-dimensional; the excess resistance (4.1) would then represent the intrinsic resistance alone.

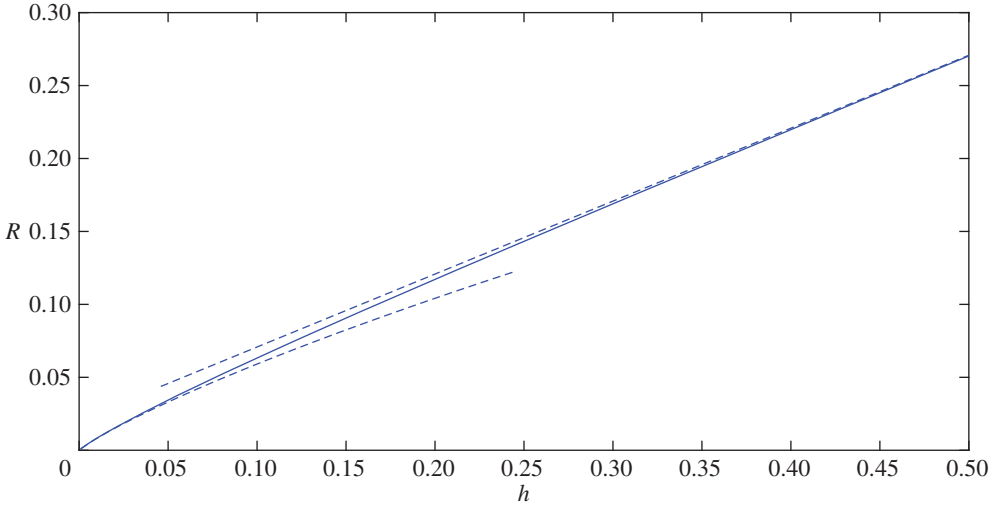


Figure 3. The dependence of the excess resistance R upon h (solid line). The dashed lines show the small-h and large-h asymptotes, (4.7) and (4.8). (Online version in colour.)

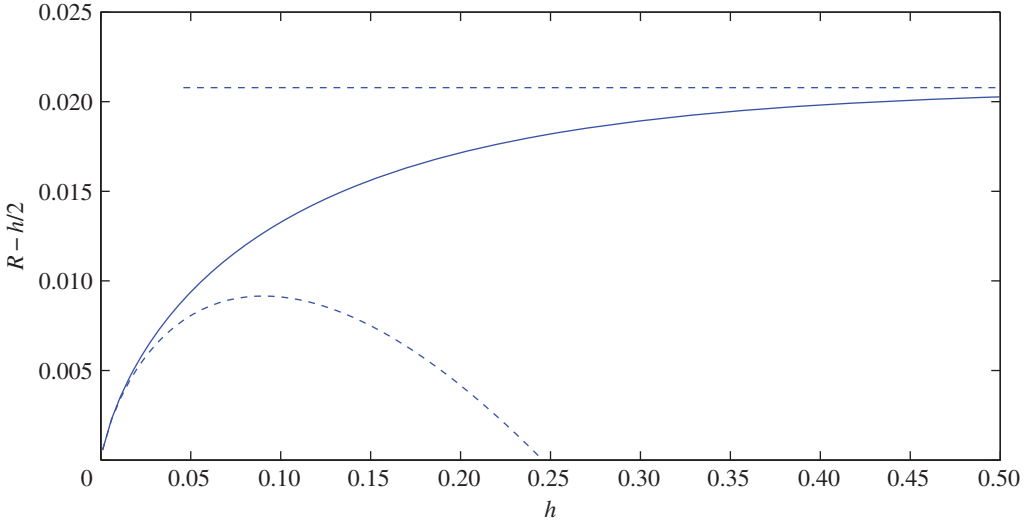


Figure 4. The results of figure 3 re-plotted to show $R - h/2$ as a function of h (solid line). The dashed lines show the small-h and large-h asymptotes, (4.7) and (4.8). (Online version in colour.)

dimensionless notation as previously. The far-field condition (2.6) is replaced by

$$\varphi \sim \frac{y}{2L} \quad \text{as } y \to \pm\infty. \tag{5.1}$$

By symmetry, the array problem is equivalent to that of a single channel of width $2L$, with a constriction of length $2h$ within which the width is reduced to 2. In this equivalent problem the far-field condition (5.1) is retained; in addition, a no-flux condition is imposed at $x = \pm L$. This equivalent problem was solved (using a Schwarz–Christoffel transformation) by Gurevich [19] and Flagg & Newman [20] in the context of added-mass calculation, and later on by Gat *et al.* [9] in the context of Hele–Shaw flows. The particular case $h = 0$ has also been studied by Dufrêche *et al.* [21].

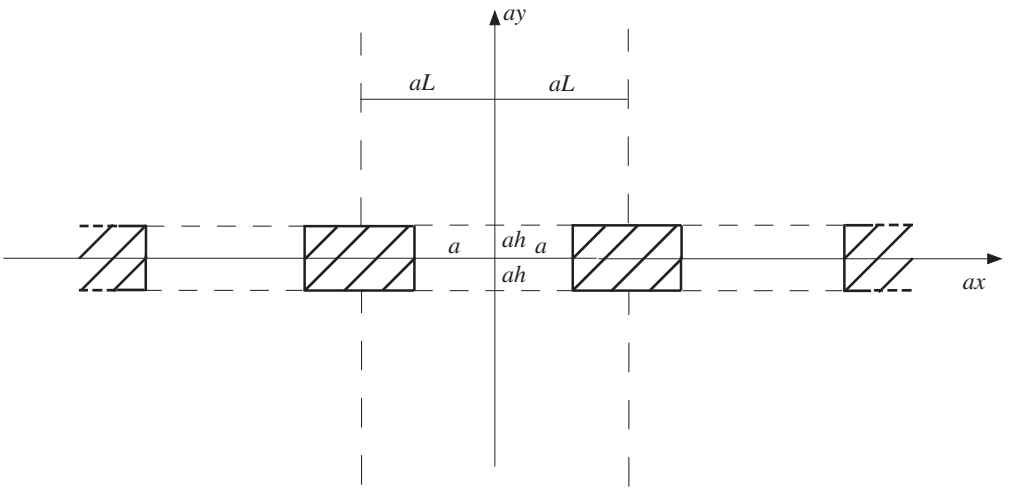


Figure 5. An array of slots with period 2*aL*.

The quantity of interest in the array problem is the excess potential difference due to the array blockage. This quantity is known as the 'blockage coefficient' in irrotational-flow problems and as the 'equivalent length' in viscous-flow problems. Despite the existence of an exact solution for the array problem, the associated equivalent length is provided as an integral, the parameters appearing therein being implicit functions of the array geometry (e.g. Gat *et al.* [9, eqn (3.16)]). There is accordingly merit in deriving closed-form approximations for the blockage coefficient in various limiting cases.

In what follows, we employ our single-slot solution towards that end. Specifically, it is expected that for large L we can solve the array problem by superposing an infinite number of single-slot solutions (cf. [22]). The associated infinite sum for the potential does not converge, but we circumvent this by considering potential differences, rather than the potential itself. Clearly, this superposition satisfies Laplace's equation, and it remains to verify that the no-flux boundary condition and the far-field condition (5.1) are satisfied.

For definiteness, we consider a channel that lies between $x = \pm L$. As each of the single-slot solutions satisfies an impermeability condition at $y = h$, their sum does so too. We therefore need to verify that impermeability is also satisfied on the vertical walls. By symmetry, it is sufficient to consider only the vertical walls $x = \pm 1$ of the zeroth slot. Now, while the field associated with the current through the nth slot (with $n \neq 0$) is not defined on these walls, its far-field behaviour is defined there. This far-field behaviour decays inversely with distance (2.3); since contributions to $\partial\varphi/\partial x$ from slots centred at $x = 2nL$ have opposite sign to those from slots centred at $x = -2nL$, the associated Leibniz series converges. The error associated with the violation of the no-flux condition on $x = \pm L$ is therefore of order $1/L$, and is thus negligible as $L \to \infty$.

It therefore remains for us to verify that the far-field condition (5.1) is satisfied. Consider the point $(0, y)$, say, with $y > 0$. For $y \gg 1$, we can employ the approximation (4.5) for the contribution of the $n = 0$ slot, giving the potential

$$\varphi = \frac{1}{\pi}\left[\ln y + \ln \frac{2}{M\sqrt{1-k^2}} + O(y^{-1})\right]. \tag{5.2}$$

Since the potential associated with this slot vanishes at $y = 0$ (see (2.7)), (5.2) also provides the contribution of this slot to the potential difference

$$\varphi(0, y) - \varphi(0, 0). \tag{5.3}$$

The far-field approximation (4.5) may clearly be used to calculate the contribution of all other slots to the potential both at the generic point $(0, y)$, for which $r^2 = y^2 + (2nL)^2$, and at the origin $(0, 0)$,

where $r^2 = (2nL)^2$; the contribution of the nth slot to the potential difference (5.3) is $(2\pi)^{-1} \ln(1 + y^2/4n^2L^2)$, and the sum of the contributions of all slots ($n \neq 0$) is

$$\frac{1}{\pi} \ln \prod_{n=1}^{\infty} \left(1 + \frac{y^2}{4n^2L^2}\right), \tag{5.4}$$

with an error of order $1/L$. Using the identity

$$\frac{\sin \pi z}{\pi z} = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right), \tag{5.5}$$

the infinite product in (5.4) is simply $(\pi y/2L)^{-1} \sinh(\pi y/2L)$. Hence when all slots (including $n = 0$) are included in the summation, the total potential difference becomes

$$\varphi(0, y) - \varphi(0, 0) = \frac{1}{\pi} \left[\ln \sinh \frac{\pi y}{2L} + \ln \frac{4L}{\pi M \sqrt{1 - k^2}} + O(y^{-1}, L^{-1}) \right]. \tag{5.6}$$

For $y \gg L$, we then obtain

$$\varphi(0, y) - \varphi(0, 0) = \frac{y}{2L} + \frac{\Delta}{2} + O(L^{-1}, \mathrm{e}^{-\pi y/L}), \tag{5.7}$$

where

$$\Delta = \frac{2}{\pi} \ln \frac{2L}{\pi M \sqrt{1 - k^2}}. \tag{5.8}$$

Since φ vanishes at $y = 0$ (by symmetry), the far-field condition (5.1) is indeed satisfied. The first term in (5.7), linear in y, represents the potential difference that would exist in the absence of the perforated plate; it is required to sustain the current density in the negative y-direction. The second term in (5.7) represents the effect of the array upon this potential difference. The associated equivalent length Δ, given by (5.8), can be expressed in terms of the slot excess resistance R, given by (4.6):

$$\Delta = 2R + \frac{2}{\pi} \ln \frac{2L}{\pi}. \tag{5.9}$$

Comparing (5.9) with the definition (3.14) for the equivalent length Δ in Gat *et al.* [9] and noting their different choice of length scale, it is readily seen that our Δ is the same as that defined by Gat *et al.* [9]. The geometric parameters used by Gat *et al.* [9], d and l, respectively, represent the ratio of the slot width to the array period and the ratio of the slot length to half of that period; expressed in our notation, $d = 1/L$ and $l = 2h/L$.

Of course, the expression for Δ given by (3.16) of Gat *et al.* [9] is valid for all d and l (or h and L), whereas our expression (5.8) is only valid for large L. We therefore expect that expression (3.16) of Gat *et al.* should agree with our (5.8) in the limit $L \gg 1$. This can be verified in two of the limiting cases considered by Gat *et al.*, $l = 2h/L \to 0$ and $l \gg 1$, for which they provide asymptotic approximations for their expression (3.16).[6] In the limit of a short constriction, $l \to 0$, Gat *et al.* [9] and Dufrêche *et al.* [21] obtained the approximation (in the present notation)[7]

$$\Delta \sim -\frac{2}{\pi} \ln \sin \frac{\pi}{2L}. \tag{5.10}$$

When considering our corresponding limit $h \ll 1$, for which $R \to 0$ (4.7), we find using (5.9) $\Delta \sim -(2/\pi) \ln(\pi/2L)$. This is indeed the large-L limit of (5.10).

[6]The other limit considered by Gat *et al.* [9], $d \gg 1$, represents an expansion of the flow channel, rather than a constriction, and is incompatible with our present analysis.

[7]There appears to be a typographical error in equation (4.1) of Gat *et al.* [9], where $2/\pi$ should be replaced by $\pi/2$.

In the limit $l = 2h/L \gg 1$ (a long constriction), Gat *et al.* [9] obtained their approximation (4.2) for Δ. For large L (i.e. small d), their approximation reduces, in our notation, to

$$\Delta = h + \frac{1}{\pi}\left(2\ln\frac{L}{4} + 2\right) + o(1). \tag{5.11}$$

The long-constriction limit corresponds to $h \gg 1$, for which (4.8) and (5.9) readily reproduce (5.11).

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